

Love AI More. Trust AI Less.

A Data-to-LaTeX Reasoning Walkthrough

Discrete Structures and Critical Thinking

Abstract

AI can help us generate explanations, code, and hypotheses. It cannot remove our obligation to check what a measurement means. This short report models a reusable workflow on concurrency data from a local-model benchmark.

1 Opening hook: a plausible answer can still be wrong

Ask the class: “If we send more requests to an AI server and the total tokens per second rises, did the GPU become faster?” An AI answer might confidently say yes. Today we test that answer. The lesson is not anti-AI: use it to suggest hypotheses, then make the evidence earn trust.

2 Claim and why it matters

Claim. Increasing the number of simultaneous requests should increase aggregate throughput until the available parallelism is saturated. The tempting shortcut “bigger GPU = faster” is not precise enough: it mixes hardware, scheduling, setup overhead, and memory placement.

3 Evidence: a reproducible table

The source is a clean derivation of 24 benchmark observations: four hardware/parallel-setting series, each tested at six batch sizes. Each cell is aggregate tokens/second followed by the overlap factor, computed as aggregate rate divided by median per-request generation rate.

Hardware	parallel limit	$N = 1$	$N = 4$	$N = 8$	$N = 32$
Tesla T4	1	69.2 / 0.49	74.1 / 0.54	73.4 / 0.53	73.4 / 0.54
Tesla T4	8	72.6 / 0.53	102.8 / 2.18	117.3 / 4.77	121.9 / 4.84
GTX 1060 3 GB	1	59.6 / 0.61	84.5 / 0.86	90.2 / 0.92	94.6 / 0.98
GTX 1060 3 GB	4	73.6 / 0.99	112.1 / 3.76	120.7 / 3.98	119.9 / 3.97

Each cell: aggregate tokens/s / overlap factor.

4 Evidence: see the shape

5 Reading the result

With parallel limit 1, throughput approaches one stream: the GTX 1060 rises from 59.6 to 94.6 tokens/s as setup overhead is hidden, but it never demonstrates four requests generating at once. With limit 8, the T4 reaches 125.3 tokens/s at $N = 16$ and then levels off. With limit 4, the GTX 1060 reaches 120.7 at $N = 8$ and then levels off. Thus the knee is near the parallel limit, not automatically at the largest batch size.

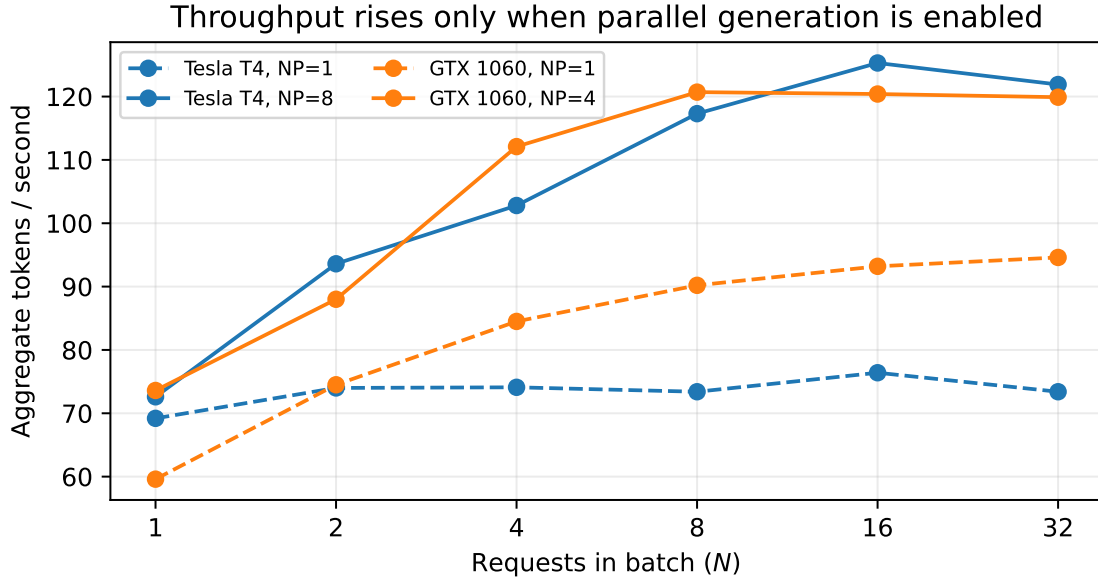


Figure 1: More requests help when genuine parallel generation is available. Dashed lines have parallel limit 1; solid lines have limits 8 or 4.

6 Discrete structures connection: relations and counting

Order the observations by $N \in \{1, 2, 4, 8, 16, 32\}$ and define a relation $N_i \preceq N_j$ when $N_i \leq N_j$. The claim predicts a nondecreasing throughput relation, but the T4 limit-8 series falls from 125.3 at 16 to 121.9 at 32: the relation is not strictly monotone. The experiment’s design is also a product-rule count: 4 series $\times$ 6 batch sizes = 24 observations. Conditioning on a parallel limit is the logical move that prevents comparing unlike cases.

7 Assumptions, caveats, and the check

- Token counts and per-request rates are roughly comparable across a series.
- The parallel limit is derived from the experiment setting; the probe recorded the server value as “auto” in some runs.
- GPU placement is an outcome: the 3 GB card’s limit-4 run was 84% GPU / 16% CPU.
- Placement and ambient load can confound a cross-machine comparison; direct generation timing would improve the study.

The check is the overlap ratio. At T4 limit 8 and $N = 16$, $125.3/25.5 \approx 4.91$, so about five requests were genuinely generating at once. At T4 limit 1 and $N = 16$, $76.4/138.3 \approx 0.55$, so the rising curve is overhead being hidden, not batching.

8 Confidence decision

Decision: medium-high confidence, bounded claim. More concurrency helps until the configured parallelism saturates in these measured settings. “Bigger GPU = faster” is not a safe universal conclusion.

9 Instructor notes and student worksheet

Suggested prompt: “Point to the claim, evidence, assumptions, check, and confidence sentence. Which one would an AI-generated answer most likely blur?” Emphasize that the overlap calculation is a proof-like check: it tests whether the data measures the proposed mechanism.

Activity (5 minutes). In pairs, choose one series. Write (1) a one-sentence claim, (2) two numbers as evidence, (3) one assumption, and (4) an overlap-ratio check. Mark the claim supported, refuted, or undecidable. Share one disagreement.

Reflection. What would you measure next to distinguish “the GPU is saturated” from “setup overhead is hidden”? Why would that measurement change your confidence?

15-minute flow. 0–2 hook and prediction; 2–4 claim/evidence workflow; 4–7 table and relation/counting connection; 7–10 plot and knee; 10–12 overlap check; 12–15 pairs activity and reflection.